Preliminary Design Presentation

Group 30



Introduction

~~Fun~~ Facts

- **44%** of U.S. road-work fatalities involved vehicle strikes (2011–2022)
- In Toronto, the average driver lost **118 h/year** to congestion in 2022, costing about **$11B** annually in productivity/opportunity.
- Work zones cause 24% of non-recurring delay; better management can cut delay **50–55%**.
- Work-zone intelligent transport systems reduced incident clearance from **45 to 25** min.



Problem Statement & Need Identification

Problem Statement

Traditional traffic pylons need manual deployment that exposes road workers and emergency responders to moving traffic. This process is labour-intensive, time-consuming, slow to adapt, and potentially hazardous.



Need Statement

There is a need for an autonomous traffic pylon system that can safely and efficiently deploy, and position traffic control devices, reducing worker exposure to traffic.

Constraints



1. Physical Design

a. Size: The robot will support a 0.5 m pylon for highway traffic control

2. User Experience

a. Worker exposure: Deploy the full pylon formation with the responder remaining outside live traffic lanes for 100% of the deployment process
b. Status Visibility: 100% of robot states are visible to the operator

3. Safety & Fail Safe Functions

a. Manual Intervention: The operator must have the ability to reposition/stop the robot(s) with a reaction time of 1 s of command
b. Fail-safe procedure: In cases of communication or power loss, the robot(s) must automatically stop

4. Autonomy & Motion

a. Obstacle response: A detected obstacle within 0.25 m must cause the robot(s) to stop or re-route
b. Positioning accuracy: Each robot should be positioned within 0.25 m of its target.
c. Autonomy: Robot(s) travels path to commanded location 100% autonomously

Objectives



1. Physical Design

a. Weight: Under 23 kg
b. Form factor: System is stackable for compact transportation
c. Weather resistance: Robot is weatherproofed for outdoor conditions

2. Mobility

a. Speed: Robot(s) reach a maximum travel speed of 5 km/h
b. Road Incline: Robot(s) can tolerate a road incline of 10°

3. Deployment & Recovery

a. Deployment time: Deploy robot(s) within ≤2 minutes of receiving the deployment command
b. Efficient recovery: Retrieves the robot(s) automatically to the operator

4. Power

a. Power endurance: Provide enough battery capacity for at least 30 minutes active deployment time

5. Budget

a. Budget: Aim to spend less than $4800

Criteria

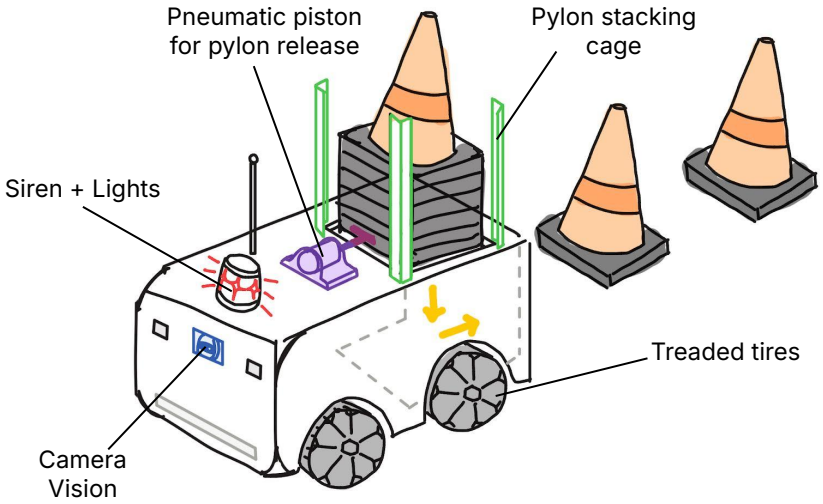




Design Criteria	Source	Description	Evaluation Metric
Physical Form Factor	Objective 1a, Objective 1b	Evaluates the compactness of the pylon system	System weight [kg]
Path and Positioning Accuracy	Constraint 4a, Constraint 4b	Evaluates the robot's ability to detect and respond to obstacles while accurately navigating to its target position	Standard Deviation from Target Position [m]
Cost	Objective 5a	Evaluates the cost of the system	CAD [$]
Terrain Traversability	Objective 1c, Objective 2b	Evaluates the ability to navigate common road surfaces, including cracks, potholes, slopes, and uneven pavement	Maximum traversable obstacle height [m] and slope [°]

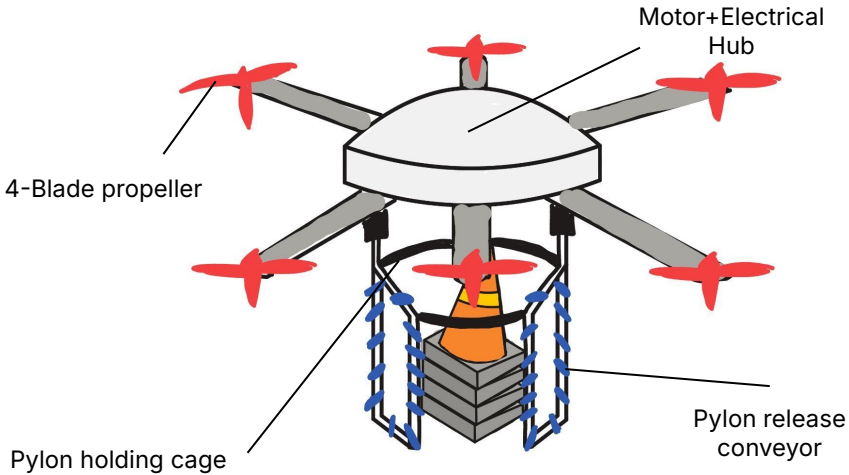
Alternate Design Solutions

Concept Design #1



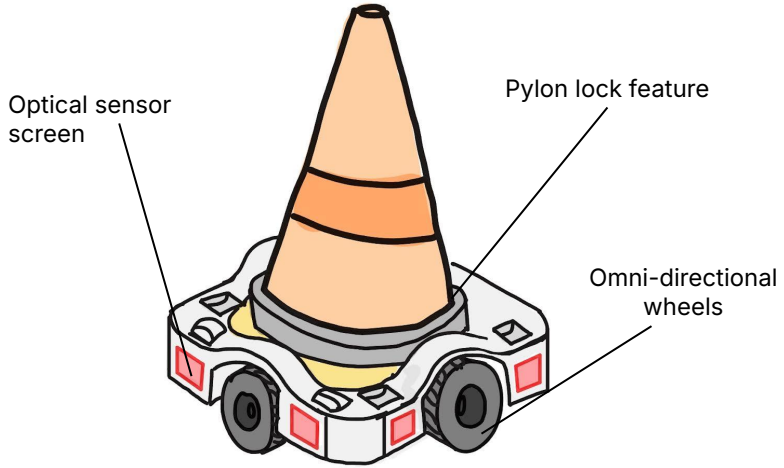
- Quick **Drop & Place** w/ pneumatic piston mechanism
- Allocated **storage** for multiple pylons
- **Sequential** pylon deployment
- Manual **remote piloting**

Concept Design #2



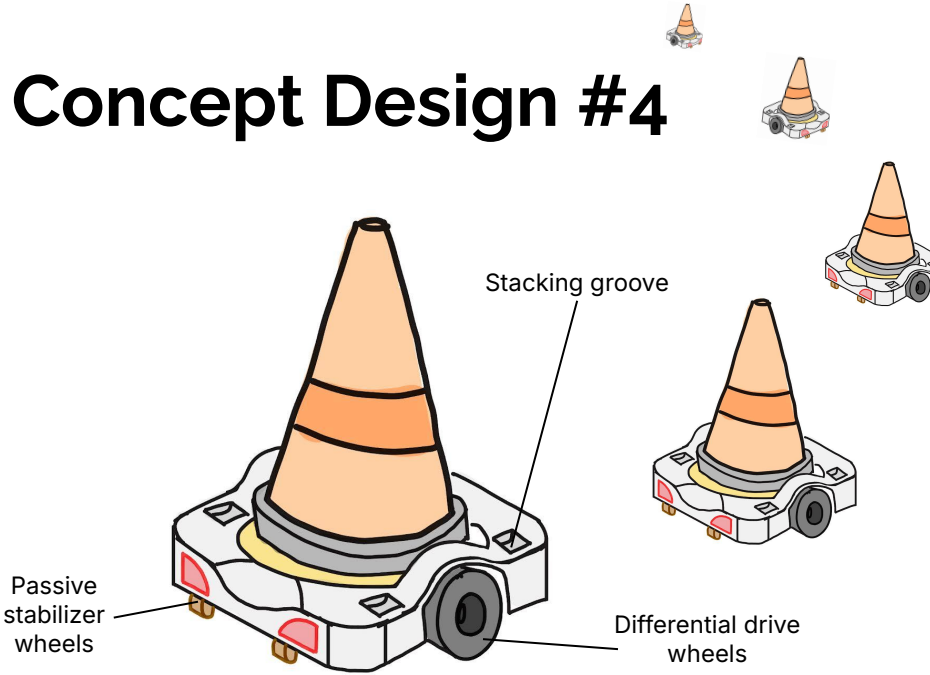
- **Drop & Place** 6-propeller drone
- Free **aerial** maneuverability
- **Sequential** pylon deployment
- Manual remote piloting

Concept Design #3



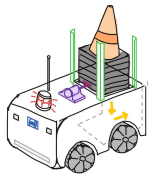
- **Autonomous** precision pylon placement
- **Swarm** pylon deployment
- Quick **attach + release** pylon locking
- **Manual remote piloting** override
- **Omni-directional** movement

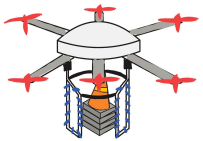
Concept Design #4

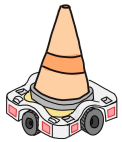


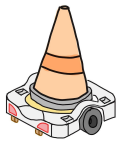
- **Autonomous** precision pylon placement
- Swarm pylon deployment
- **Stackability** for smaller storage footprint
- **Differential drive**

Design Selection Matrix









Criteria	*Physical Form Factor*	*Path and Positioning Accuracy*	*Cost*	*Terrain Traversability*	***Total***
Weights	6	10	5	8	-
Design #1	3	6	4	6	146
Design #2	1	7	2	7	142
Design #3	10	9	6	4	212
Design #4	10	8	6	8	234



Patent Review

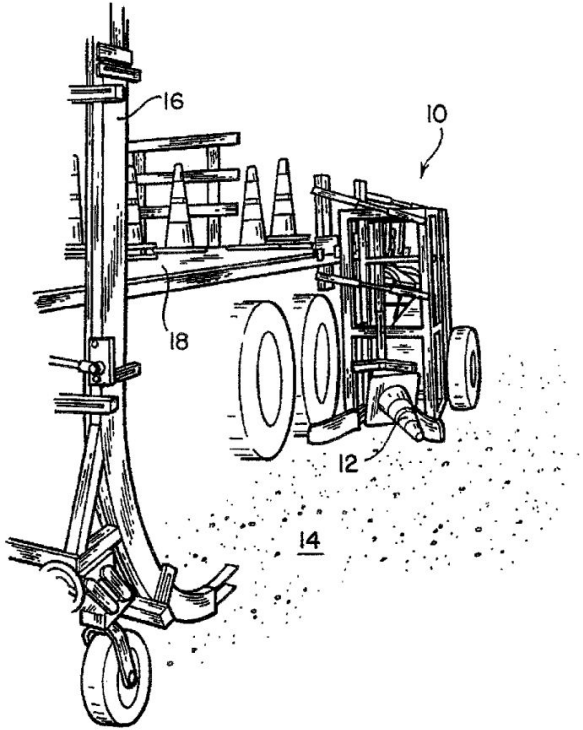


Traffic Cone Retrieving Device for Removing Traffic Cones from Roadway Surfaces (CA2441457A1)

Existing approach: A vehicle-mounted mechanism engages cones, raises them, and stores them in a nested stack.

Implications for our design:

- Retrieval and compact storage are established design considerations
- **Opportunity:** Develop autonomous recovery for independently mobile pylons, without requiring a vehicle-mounted recovery mechanism

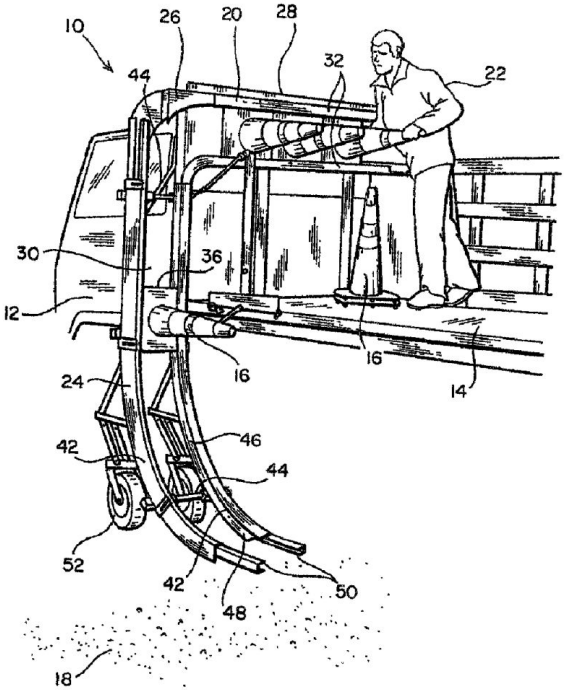


Traffic Cone Setting Devices for Depositing Traffic Cones onto Roadway Surfaces (CA2441452A1)

Existing approach: Vehicle-mounted magazine and chute dispense cones at predetermined intervals, potentially using timed automatic control.

Implications for our design:

- Automated, repeatable cone spacing is established prior art
- **Opportunity:** Enable mobile pylons to navigate to user-defined positions and form flexible boundaries, without requiring a vehicle-mounted dispensing mechanism



A Traffic Cones and Traffic Cone Lanterns Placement and Collection System and a Method (EP3643838B1)

Existing approach: Vehicle-mounted robotic arms use camera sensing and optional radar to place and collect cones while the vehicle moves.

Implications for our design:

- Vehicle-mounted robotic cone handling is established prior art
- **Opportunity:** Explore distributed deployment, flexible formations, and autonomous recovery without relying on robotic arms mounted to the vehicle

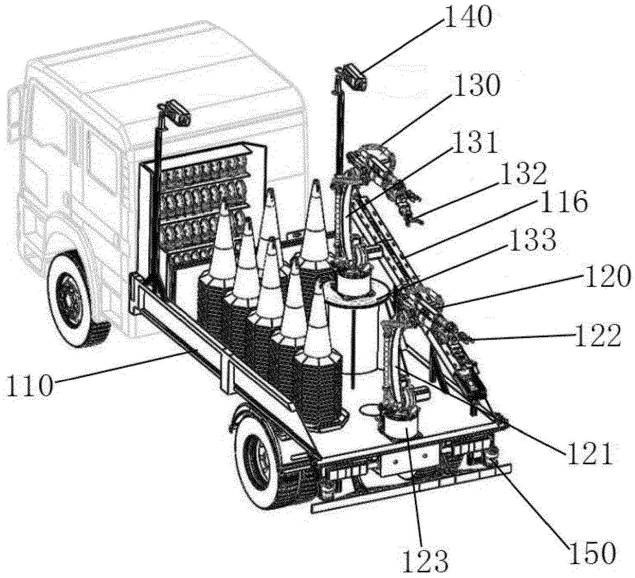
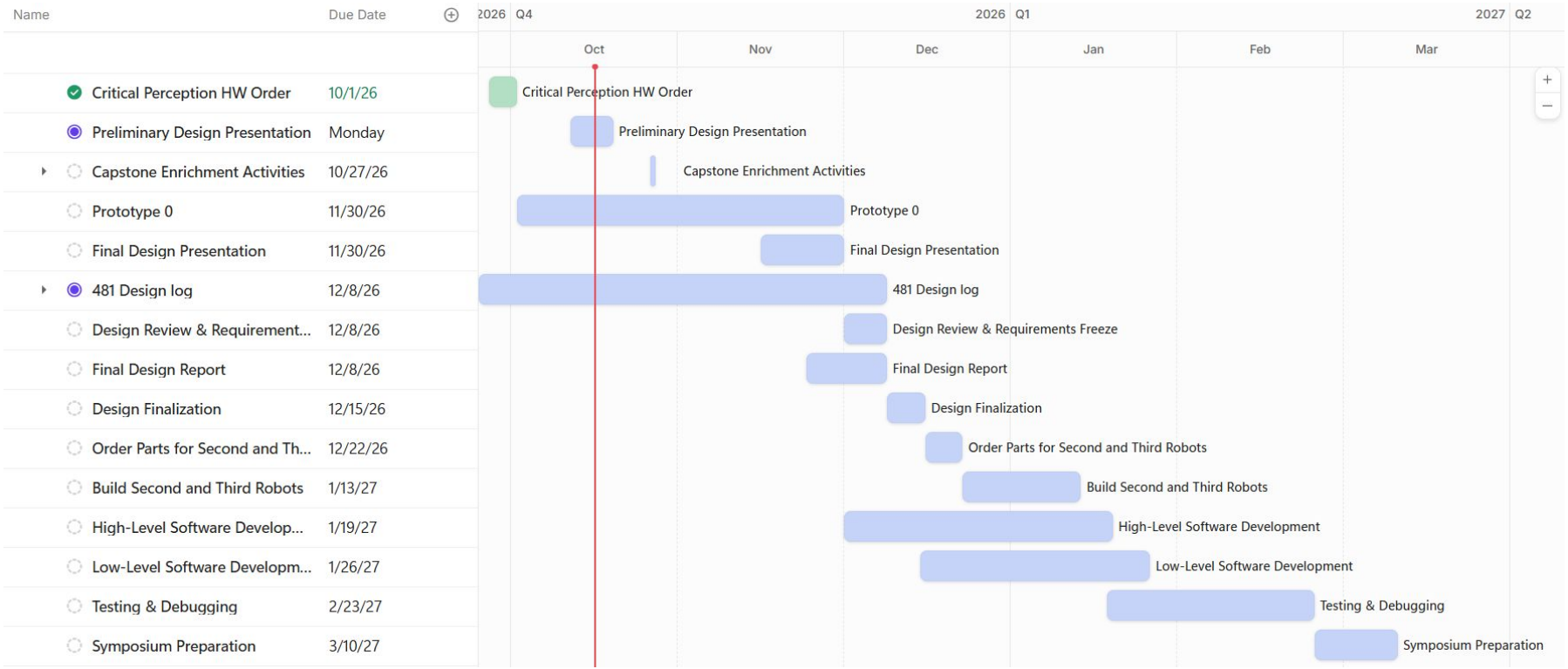


FIG. 1

Project Timeline

Our project timeline leading up to symposium has been laid out using ClickUp



References

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- https://ops.fhwa.dot.gov/publications/fhwahop19062/fhwahop19062.pdf
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